

Safely repurposing existing pipeline-infrastructure for CO₂ transport – Key issues to be addressed

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ABSTRACT:

In the context of carbon capture and storage (CCS) or utilization (CCUS), offshore pipelines are foreseen to play an important role for the transport of large quantities of carbon dioxide (CO₂) from location of sources to location of sinks/consumers. There is currently a strong interest within the industry to explore the possibility for repurposing of existing pipeline infrastructure to leverage on existing CAPEX investments. Even though this may seem attractive from a cost perspective, the technical, safety and financial risks need to be acknowledged and addressed accordingly. Several decades of industry experience exists for onshore CO₂ pipelines, mainly for enhanced oil recovery (EOR), however, the experience is rather limited for offshore pipelines [1]. Change of product will per applicable pipeline design codes require a re-qualification to ensure that the new premises for change in operation is properly assessed and confirmed acceptable with regards to pipeline safety, operability, and transport capacity. Per DNV-ST-F101, Sec.11.E [2], re-qualification involves a re-assessment of the pipeline system for the new premises. A work process for re-qualification of pipelines to operation with CO₂ is provided in DNV-RP-F104 'Design and Operation of CO₂ pipelines' [3]. This paper addresses the key challenges, considerations and requirements relevant for repurposing of subsea oil & gas pipelines for transporting CO₂, with main focus on pipeline structural integrity.

1. INTRODUCTION

The potential for repurposing of existing onshore and offshore pipeline infrastructure for transporting CO₂ has over the last decade been given increased attention. Both regarding cost and environmental footprint of building new dedicated pipelines, repurposing is considered a potential attractive option. The topic is addressed in several previous research studies, such as Rabindran et al [4]. In a recent techno-economical study performed by Carbon limits and DNV [5], a high-level screening of the European oil & gas pipeline network¹ was performed to assess repurposing potential either for transporting CO₂ or hydrogen gas. The study concludes on a significant potential for CO₂, however also that the re-use potential will depend on several factors such as location, pipeline route, product composition, operating condition (dense or liquid phase), physical condition and other factors. There are several key criteria that need to be considered carefully, and checks that need to be made, before approving a piece of onshore or offshore infrastructure for re-use. Implicitly the feasibility of repurposing of a specific pipeline needs to be confirmed and documented through a re-qualification process to confirm acceptable pipeline integrity as well as transport capacity. It is the intention of this paper to describe a methodology and to address key considerations specifically needed for re-qualifying oil & gas pipelines for transport of CO₂ in line with guidance provided in DNV-RP-F104.

¹ The study included data collection from 65 pipeline operators, covering a total of 58,000 km of onshore and offshore pipelines.

2. THE VALUE OF INDUSTRY STANDARDS & GUIDELINES

Industry standards, recommended practices and guidelines provide requirements, specifications, guidelines, and characteristics that can be used consistently to ensure that materials, products, processes, and services are fit for their purpose. Further, they provide guidance for the safe management of pipeline infrastructure - both for new design and re-use. Ultimately, they reflect industry experience and are often results of joint industry projects which establishes trust and confidence between stakeholders, authorities, and society. For pipeline transport of CO₂, there are existing design codes that provides design criteria and guidance specifically for CO₂ transport such as e.g. ASME B31.4 [6], DNV-RP-F104 [3] and ISO 27913 [7]. Figure 1 provides a high-level overview of existing codes across the CCS value chain within the ISO and DNV regime.

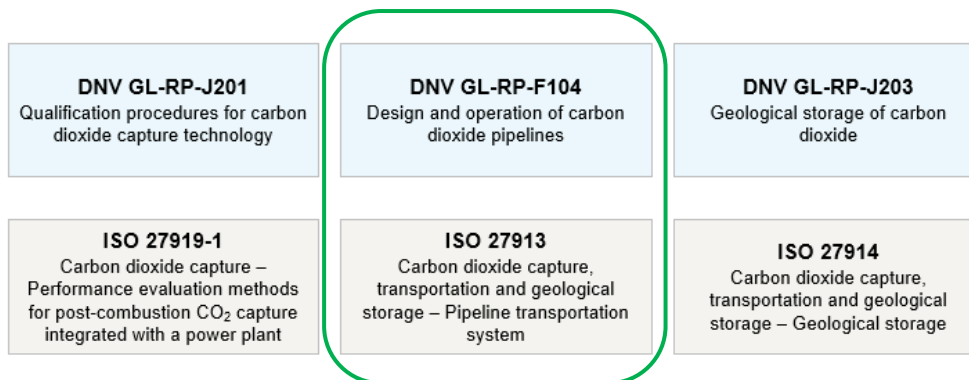


Figure 1: Overview of existing DNV and ISO codes related to CCS

3. KEY CONSIDERATIONS OF REPURPOSING PIPELINES FOR CO₂ TRANSPORT

In the context of carbon capture, the CO₂ may come from various sources and be captured by different techniques leading to variation in product (CO₂) composition. Within the industry, there are specifications stating minimum 95% CO₂ in the composition, where the remaining 5% is typically represented by hydrocarbons, nitrogen and other non-condensable, and with traces of sulfur, oxygen, glycols, and water [8]. The CCS industry is currently requiring a conservative composition close to 99% CO₂, driven by limitations on acceptable impurities across the value chain from capture to storage or utilization. It is foreseen that this strict composition will be challenged in the future, hence this should be accounted for when considering repurposing of pipelines for CO₂ transport. Further, industry experience shows that the corrosion rates, and possibly the impact on fracture control evaluations, are strongly affected by the type of impurities, combination of impurities and concentration of impurities [9].

The first obvious challenge relates to risk of internal corrosion for C-Mn steel due to the high corrosion rates due to CO₂, hence the requirement to avoid free water across the pipeline system and foreseeable operating conditions. This challenge is also further complicated by the presence of sulfur regarding sour service conditions. Corrosion risk assessment also needs to consider the additional effects of traces of oxygen in the composition.

The second main challenge is related to what extent the pipe could have sufficient toughness to arrest a running ductile fracture when operated in dense phase, which is also affected by the composition of the CO₂ and the material properties of the pipeline. There is now an increasing body of evidence regarding the more challenging situations with respect to running ductile fracture in dense phase CO₂ pipelines. It has become clear that the original Battelle Two Curve Method is non-conservative when

applied to dense phase CO₂ pipelines [10]. Recent large-scale tests point to potential requirements for high toughness, with CVN energy above 300 J in some cases, to arrest running ductile fracture (see [11] for a summary). Such high CVN energy values could be challenging to document for older CMn-pipelines and could point to the need for considering alternative solutions, e.g. the use of crack arrestors as commonly used for onshore CO₂ pipelines. For offshore pipelines, retrofitting crack arrestors may however be challenging and costly.

There has also been raised a concern as to whether smaller leaks in CO₂ pipeline systems could lead to more severe cooling, which again could increase the risk of brittle fracture. The importance of this issue is, however, not fully understood [12].

From a more general perspective there are also potential challenges that needs to be handled related to change of flow direction, not only related to one-way equipment but also due to the change of pressure and temperature profiles with regards to pipeline load conditions.

Further, interpretation of existing regulations and potential change of location class (or safety zones) also needs to be evaluated as part of the re-qualification process.

4. HOW TO SAFELY REPURPOSE A PIPELINE SYSTEM FOR CO₂ TRANSPORT

4.1 General

DNV-RP-F116 'Integrity management of submarine pipeline systems' [13] provides recommendations on how to manage integrity of submarine pipeline systems for the intended service life (Figure 2). The main objectives are to:

- Provide guidelines to ensure that the operation of pipeline systems is safe and conducted with due regard to public safety, environment, and properties
- Provide guidance on how to comply with the requirements given in DNV standard DNV-ST-F101
- Serve as a guideline for operators and suppliers.

Pipelines transporting CO₂ are not specifically addressed in DNV-RP-F116 but gives references to DNV RP-F104 [3] covering pipelines for CO₂ transport specifically. Generally, when a pipeline is considered used for a purpose other than it was originally intended, or by not fulfilling the original design criteria, a need for re-qualification is triggered [2].

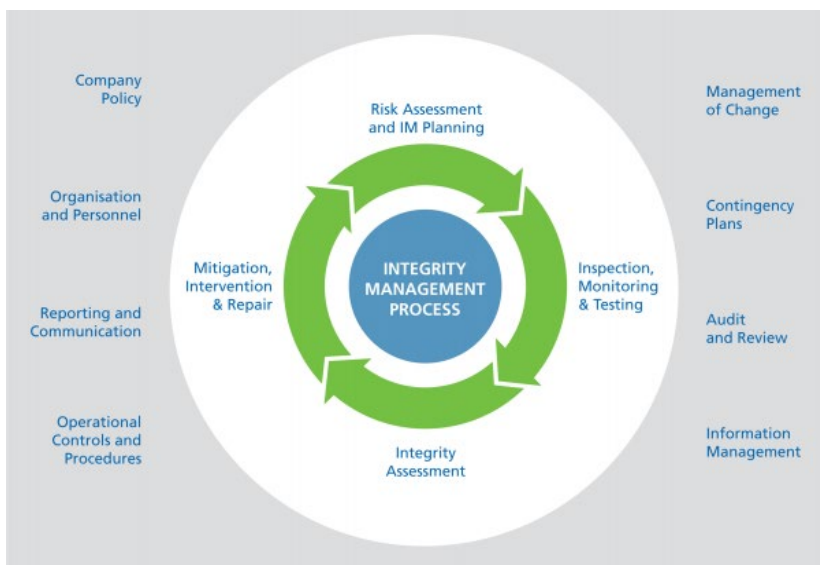


Figure 2 Integrity management system

4.2 An outline of the various steps in a re-qualification process

A re-qualification process is initiated by a decision to evaluate the possibility of bringing an existing pipeline system or parts thereof into operation with transport of CO₂. This change is primarily identified through change of medium in the pipeline from the original design. As a general rule, the re-qualification shall comply with the same requirements regarding safety and operation as for a pipeline designed specifically for transportation of CO₂. The re-qualification process is recognized through the process given in Figure 3 and briefly described through the steps 1-9 [3] below.

Step 1 - Initiation: A decision is taken to evaluate an existing pipeline system for the purpose of CO₂ transport. An essential part is to establish the original design basis and any later modifications, operation parameters, operational history and battery limits extracted from the DFI résumés, and further to identify key elements within the battery limits and identify the materials selection as well as the pressure rating of the pipeline system. This is key information that needs to be screened as part of the re-assessment part (step 7), i.e. basis for gap analysis for requirements for CO₂ operation. A risk assessment should be carried out and a project risk register developed which identifies the risks associated with changing the type of product to be transported from existing to in dense phase or gas phase. The base requirements for the re-qualified pipeline should be established. This includes aspects such as 1) capacity (how much CO₂ to be transported), 2) delivery pressure, 3) CO₂ state and composition to be transported which will influence the later integrity and safety studies.

From recent studies, often seen gaps related to this step is lack of;

- Original design basis / Design Fabrication Installation (DFI)
- Material certificates

Step 2 – Integrity assessment: As a starting point, the current integrity of the pipeline system shall be addressed through assessment of the technical condition prior to changing to CO₂ operation. Historical information of how the system has been operated compared with the requirements for operation should be assessed and documented. The information from an integrity management process/program as available may be beneficial as support, but this may not necessarily provide all information needed (e.g. lack of inspections documenting current status, lack of operation data, lack of design documents). Identification of material selections as well as the pressure rating and the current condition of the pipeline system are key information that shall be screened as part of the integrity assessment.

Step 3 – Hydraulic analysis and flow assurance

Changing existing product to CO₂ may have implications for the pipeline transport capacity and load conditions. A flow analysis should be performed to identify feasibility with regards to transport capacity and corresponding pressure and temperature distribution along the route. A high-level hydraulic analysis, comparing the key operational parameters when operating with current product

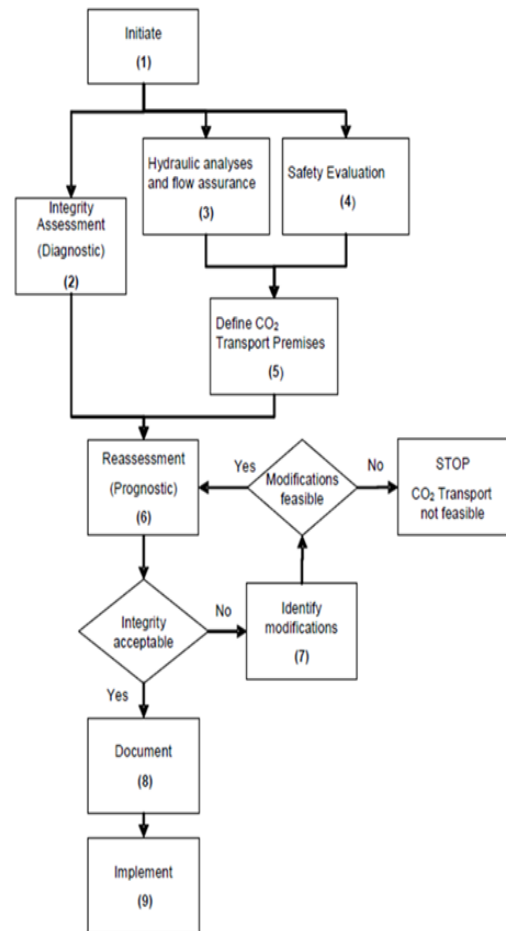


Figure 3 Re-qualification process for pipeline system change into CO₂ transport

and CO₂ respectively shall be performed as input to the pipeline load conditions. A representative set of operating conditions shall be covered to establish pressure (steady state and transient operation) and temperature profiles. Requirement for modifications of the pipeline system should be identified and evaluated. These aspects are further discussed in Section 5.

Step 4 – Safety Evaluations

Safety evaluations should be performed, addressing the implications of a change of product. The system as it is designed should be evaluated toward the specific safety requirement for CO₂ pipelines. Need for modifications of the pipeline system should be identified and evaluated, e.g. additional block valves, upgrade of leak detection system etc. The suitability of components / replacements of valves and gaskets not suitable for CO₂ should also be evaluated. Attention shall be given to accidental release scenarios, and the effects on consequence radius due to the change of product. These aspects are further discussed in Section 6.

Step 5 – Define CO₂ transport premises

Premises for CO₂ transport should be defined incorporating the results from the hydraulic analysis (3) as well as safety evaluation (4). These aspects will define functional as well as system requirements that may deviate from the original design conditions. Requirements to the system should be identified and included as part of a complete premise for the re-qualification. This includes stating design / incidental pressure, receiving pressure etc.

Step 6 – Reassessment

Based on the input from the integrity assessment (2) and the CO₂ transport premises (5), the integrity should be evaluated for the new premises. Parts of the system that is identified not to be compliant with the integrity requirements will require design modifications and reiteration on the re-assessment. In some cases, the required modifications are not feasible e.g. from a cost-benefit standpoint, leading to termination of the re-qualification process.

Assessment of additional Failure Mechanisms: Additional failure mechanisms relating to the changed operational conditions shall be identified, considered, and evaluated (e.g. change of service from natural gas (NG) to CO₂ in gas or dense phase with various compositions). Key areas to be addresses are typically:

- Compliance with existing material properties
- Load conditions: Re-assessment of the pipeline system, specifically addressing the implication of new premises on pipeline load conditions. The assessment shall be performed comparing the requirement in the original design code and with specific requirements for the system in question.
- The change in operational conditions as:
 - Maximum allowable pressure (MAOP)
 - Changes in product composition and weight
 - Flow velocities (related to transport capacity)
 - Design factors
 - Design temperature
 - Increase/decrease in pressure cycling (fatigue).

Risk register: The risk register established in (1) shall be updated and any risks associated with the pipeline including upstream and downstream battery limits identified. The risk assessment shall confirm that these changes are appropriately addressed in relevant management and replacement plans for the pipeline.

Comparison of original and new design standards: A gap-analysis to identify gaps between the original design standard and applicable current design requirements and CO₂ specific guidelines or

recommended practices should be performed. Re-assessment of the pipeline system shall be performed, specifically addressing the implication of new premises on pipeline load conditions.

Reference is also given to Section 7 for more details around design aspects to be considered.

Step 7 – Modifications

Modification alternatives should be evaluated with respect to feasibility, safety, and integrity. A re-assessment of the modification alternative will be performed through documenting the integrity status. This activity shall cover identification and description of possible mitigations or modifications to ensure safe operation with CO₂, considering:

- Acceptable levels of impurities
- Acceptable operating envelope
- Other modifications/mitigations for improving load condition for the pipeline
- Material testing to document (original) material properties when information is lacking
- Any commissioning and Ready for Operation activities related to change of product and first fill

Step 8 – Document

Documentation of the re-qualification process as well as update of system documentation, drawing, equipment lists, and operating procedures is required to ensure that the re-purposed pipeline can be safely and effectively operated.

Step 9 – Implement

Implementation of changes to the system should be performed prior to transition of the system into the new operational mode for CO₂.

5. HYDRAULIC ANALYSIS AND FLOW ASSURANCE (STEP 3)

For existing pipelines purposely built for transporting CO₂, the pipeline design and operating envelope is in most cases adapted to transport the product in liquid (dense) phase. This is due to the favorable combination of high product density and low viscosity for the liquid phase, allowing for optimization of both installation (i.e. reduced pipeline diameter) and operational cost. Figure 4 shows a typical phase envelope for pure CO₂ and for a 'synthetic' composition containing 97%CO₂ and 3%N₂. The 'synthetic' composition is included to demonstrate typical effects of other impurities, which may vary depending on the source of the CO₂ captured and location in the value chain.

Operation in liquid phase requires minimum pipeline pressure rating in the range of 80 to 100 bar, depending on stream composition and range in operating temperature, ref. Figure 3. For offshore pipelines, the pressure rating will in many cases be higher than 100 bar, i.e. sufficient to enable operation in liquid (dense) phase. Offshore gas pipelines typically have a design pressure in the range of 150 bar. It should be acknowledged that in most cases it is not possible to re-qualify a pipeline to higher pressure rating, considering that pipelines are normally optimized on design loads, including pressure containment. Hence, with regards to determining transport capacity, the pipeline pressure rating should first be identified to conclude feasibility of operation in liquid phase, or whether operation in gas phase at lower pressures is the only viable option.

For typical onshore oil & gas pipelines, pressure rating is typically less than 80 bar and the pipeline would normally be operated in close balance with ambient temperature, either exposed above ground or buried. Pipelines with pressure rating less than 80 bar may still be feasible for transporting CO₂ in gas phase, however the limitation in upper operating pressure for the avoidance of two-phase flow condition needs to be acknowledged. Conservatively the maximum operating pressure for gas phase

transport may be determined by the saturation pressure at the minimum operating temperature of the line, i.e. the pressure at which a liquid phase will start to drop-out (condensation). Minimum operating temperature will typically be governed by a combination of inlet temperature and thermo-hydraulic response of the pipeline from inlet to outlet, including the effects of pipeline insulation and variations in ambient temperature. To confirm acceptable margin to two-phase flow, hydraulic simulations should be performed with appropriate tools to determine relevant range in pressure and temperature profile along the pipeline route for the actual product composition. As an example, if the minimum ambient soil temperature for a buried pipeline is 10°C, the maximum operating pressure in gas phase needs to be limited to approximately 45 bar, depending on product composition. Combined with the pipeline size, length and minimum required arrival pressure, this limitation will have direct implications for the pipeline transport capacity.

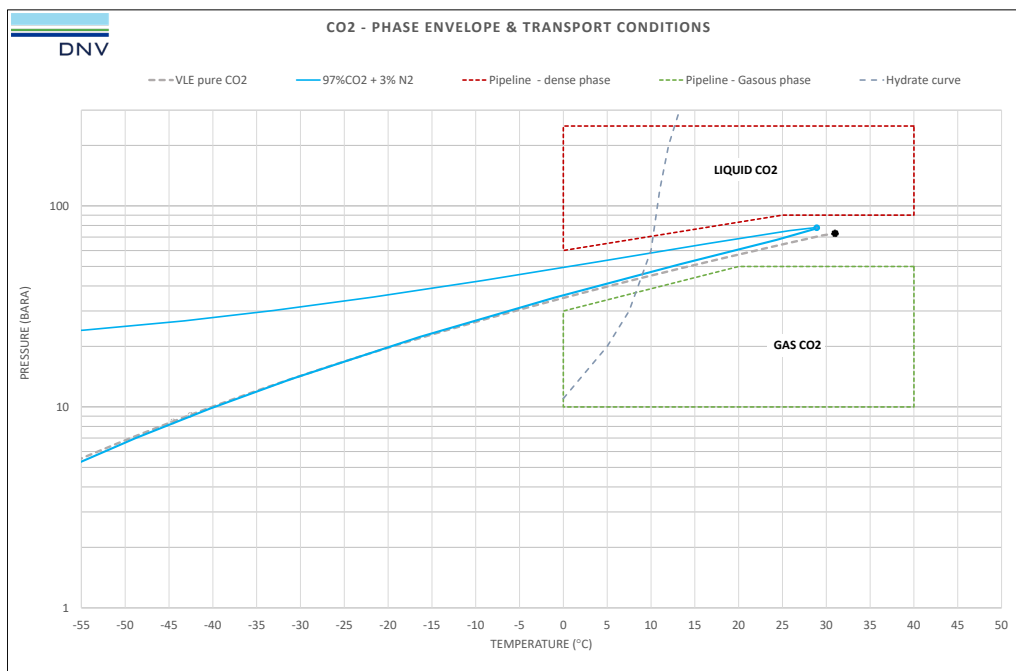


Figure 4 Phase envelope for pure CO₂ and composition of 97%CO₂+3%N₂; for illustration; typical operating envelope for dense and gas operation indicated.

Regarding flow assurance, the main concern is to ensure sufficient control on product water content to prevent drop-out of free water. In this context it should be acknowledged that water solubility varies significantly between liquid and gas phase and may also be affected by the presence of other impurities in the product [14]. Also the potential for free water drop-out in upset conditions, including off-spec product should be considered. For operation in gas phase, hydrate will typically form below 10°C, and typically below 10 to 15°C in dense phase, ref. Figure 3. At temperature below 0°C, free water will turn to ice. Hydrate risk should be mitigated indirectly by preventing free water drop-out, implicitly by controlling composition water content.

In addition to the assessment of transport capacity, temperature and pressure profiles should be established from thermo-hydraulic simulations for a representative range of operating conditions to provide input to the new pipeline load conditions. The simulation cases should typically include design cases, normal operation, turn-down case(s), line-packing, shut-down and re-start. Pipeline depressurization (rare event) should also be simulated to document that the pipeline can be safely depressurized within reasonable time and within the specified design conditions such as minimum design temperature. Repurposing existing oil & gas pipelines for transporting CO₂ will in many cases

require reversing the flow direction which will have implications for pressure and temperature profiles. Any 'one-way' equipment or functions needs to be identified and assessed. Booster or pressure reductions stations may need to be bypassed or reconfigured and non-return valves etc. may need to be removed or locked open.

6. SAFETY EVALUATIONS (STEP 4)

Safe operation of CO₂ pipelines has been demonstrated through a number of pipeline projects related to enhanced oil recovery (EOR) and CCS [1]. Most of these onshore pipelines are designed for operation in dense phase. There is currently no evidence to support that the failure frequencies for CO₂ pipelines should be higher than for comparable oil & gas pipelines [15]. However, it should be acknowledged that these pipelines are purpose built for transporting CO₂, and that special measures both regarding design and operation are implemented to manage integrity and mitigate major accident scenarios. It should also be acknowledged that most of the existing CO₂ pipelines run through remote areas with limited interference by and potential for exposing 3rd party. To lean on the current experience from onshore CO₂ pipelines, it therefore needs to be evaluated whether the relevant safeguards can be achieved and is also sufficient for existing oil & gas pipelines repurposed for CO₂ in more highly populated areas, alternatively whether this can be mitigated by other safeguards.

The safety evaluations for operation with CO₂ should be performed on the back of the hydraulic analysis to ensure correct assessment of the hazard potential, both considering normal planned operations and accidental release scenarios. Due to the significant difference in pipeline inventory between CO₂ pipelines operated in dense and gas phase the corresponding differences in consequence scenarios for pipeline failures should be acknowledged regarding release rates and duration.

7. RE-ASSESSMENT (STEP 6)

7.1 General

This section outlines some key material aspects and aspects in design of CO₂ pipelines.

7.2 Key material aspects for pipeline systems

Steel material:

As outlined in Section 3, there are materials aspects related to CO₂ transport that needs special considerations when assessing pipelines for such use. These are related to potentially increased degradation of the steel pipeline due to:

- Internal corrosion in presence of water
- Embrittlement in case of CO₂ compositions with certain impurities, e.g. H₂S
- Embrittlement in case of low steel temperatures in relation to small leaks

There are also scenarios where the CO₂ could lead to higher loading of the pipeline, and thus increased requirements to material properties, which especially is related to:

- Toughness requirements to arrest running fracture in CO₂ pipelines.

These aspects must be carefully considered when performing the re-assessment of pipeline for use in CO₂ transport.

Non-metallic materials:

Dense phase CO₂ behaves as solvent to certain non-metallic materials such as elastomer seals and gaskets. With respect to polymeric materials elastomers, both swelling and explosive decompression damage needs to be considered. Swelling of the elastomers is attributed to the solubility/diffusion of

the CO₂ into the bulk material. Explosive decompression may occur when system pressure is rapidly decreased and the gases that have permeated into the elastomers expand. Candidate materials also need to be qualified for the potential low temperature conditions that may occur during e.g. the pipeline depressurization scenario. For offshore pipelines any associated equipment such as valves will typically have primary metal-to-metal seals, still the risk of exposing non-metallic seals needs consideration.

Natural gas pipelines with internal flow coatings may run a risk of coating detachment from the base pipe material in a potential low temperature condition associated with rapid pipeline depressurization. Compatibility of internal coating with CO₂ should therefore be checked, even though the internal coating is not intended to provide any mitigation for internal corrosion; the concern relates mainly to potential coating debris and internal roughness of the pipe.

7.3 Design aspects

The same limit states as for other pipelines apply for pipelines transporting CO₂. However, due to different characteristics of the CO₂ compared to e.g. natural gas, a pipeline designed to transport CO₂ including required intervention may not be identical to a pipeline transporting natural gas, even for cases when the design pressure and diameter are identical.

Internal corrosion: If free water is present in the CO₂ flow, this could lead to significant corrosion rates and selecting a corrosion allowance to mitigate this will not be sufficient. Therefore, both strict control of the containments in the flow and drying of the flow is essential for transport of CO₂ to prevent internal corrosion. Corrosion resistance alloys may be selected for new pipelines but comes with an increased cost, and this is obviously not an option for repurpose of existing carbon steel pipelines. There could also be an issue with corrosion products being carried into sequestration wells or blocking valves. If corrosion is expected, then filtration before the CO₂ goes subsurface could e.g. be needed.

Fluid category and safety zones: The recent revisions of ISO 13623 [16] and DNV-ST-F101 [2] have moved CO₂ into a more severe fluid category compared to earlier revisions of the codes. The fluid category for CO₂ is now the same as for natural gas, and hence the design factor for pressure containment for a pipeline conveying CO₂ and natural gas is the same, provided that the population density (number of people) within the consequence zone is the same. Depending on the volumes in the pipelines, the consequence zones for CO₂ pipelines may be larger as the CO₂ is spreading along ground and not up in the air. If the consequence zone is increased this may give a larger/stricter location class, which in turn will result in a reduced pressure containment design factor which again will result in a reduced design pressure. This might be relevant for re-qualification of pipelines to CO₂ transport, particularly with regards to feasibility of dense phase operation.

Running ductile fracture: Full scale fracture arrest tests have revealed that pipelines conveying CO₂ are more vulnerable to running ductile fractures than natural gas pipelines, i.e. the design approaches applied for natural gas pipelines are non-conservative for CO₂ pipelines. Hence, a pipeline transporting CO₂ is expected to require a higher fracture toughness and/or thicker wall thickness or having restrictions with respect to CO₂ composition and temperature to control the saturation pressure which governs the driving force for running ductile fractures. For re-qualification it is not possible to increase fracture toughness or wall thickness, so in case insufficient characteristics an alternative is to control the CO₂ composition and temperature to arrive at acceptable saturation pressure, or to mitigate fracture arrest by e.g. fracture arrestors where required and possible.

Brittle fracture: If there is a small leak in a CO₂ pipeline in combination with significant stresses in the area, initiation of brittle fracture is a possible scenario. A detailed analysis of this issue is currently challenging. Beneficial information would be if existing materials certificates point to good low

temperature toughness (e.g. low CVN energy transition temperature). Further, effort to minimize the risk of leaks in critical regions would also be of benefit to reduce the likelihood of such events.

Reversed flow direction: Pipelines being re-qualified for CO₂ transport (in case of storage) may often change inlet to outlet (and vice versa). The original design of the pipeline has typically accounted for expansion forces due to pressure and in particular temperature, to allow safe operation of the pipeline. Since the temperature loading is most pronounced at the inlet end of the pipeline, typically little or no considerations have been placed to the outlet end of the pipeline due to limited expansion forces. However, when inlet and outlet ends are changed, expansion forces may be relevant at the (new) inlet end. This will require that the expansion potential and forces are evaluated and may lead to additional intervention to accommodate the increased expansion forces, by e.g. increased burial to avoid upheaval buckling or by allowing the pipeline to expand in a controlled manner. Additional challenges may be limitations at end terminations or structures in this part of the pipeline.

Effects of weight and mass of fluid: The density of the CO₂ in dense phase could be in the range of 10 times higher than the density of the natural gas, while the density of CO₂ in gas phase is more similar to natural gas. Hence, if CO₂ is transported in gas phase the weight and mass of the pipeline may not change significantly, however, for dense phase CO₂ the weight may increase significantly (in particular for submerged pipelines), and also the mass will increase. The latter is closely linked to e.g. whether the considered pipeline has weight coating or not. If the weight of the pipeline increases, this may decrease the acceptable free span lengths as the bending of the pipeline increases on supported shoulders, and additional mitigation may be required. Further, an increased mass will change the natural frequencies of e.g. free spans and hence the acceptable free span length may be shorter due to changes in the fatigue/fracture loading response, and additional mitigation may be required. For fully supported pipelines (i.e. pipelines without free spans), the effect of changes in mass and weight may have moderate effect. The dynamic loads from e.g. waves and current may also change due to changes in weight and mass. Changes in weight will also influence on-bottom stability.

Third party loads: Loads from third party are generally not changed when changing from natural gas to CO₂ transport, however, if weight and mass of the pipeline is changed this may change the response of the pipeline. Another aspect is that the level of third-party activity at the time of the original design may be different than the relevant activity as per today, implying that the loads from third party activity may be different. An example of increased 3rd party activity could be increased bottom trawling (offshore) or increased/new building activity (onshore).

8. SUMMARY AND CONCLUSION

This paper has discussed a general approach for re-qualification of pipelines for CO₂ transport. There are several key criteria that need to be considered carefully, and checks that need to be made, before approving a piece of onshore or offshore infrastructure for re-use. Implicitly the feasibility of repurposing of a specific pipeline needs to be confirmed and documented through a re-qualification process to ensure acceptable integrity, safety as well as transport capacity. General codes and recommended practices that lay out requirements and guidelines for design and operation of CO₂ pipelines are already available and can be used as basis for re-qualification of CO₂ pipelines. It is foreseen that these standards will evolve with the CCS industry to incorporate latest research.

In this paper, the various steps in the re-qualification process in DNV-RP-F104 has been outlined. There are however aspects that could benefit from further research, e.g. formulation of requirements for running ductile fracture and eventual environmental embrittlement. Impurities in captured CO₂ affect critical pressure, critical temperature, and phase behavior, which may affect pipeline materials and design parameters. Other challenges related to repurposing of pipelines may be general lack of design

and construction documentation for older pipelines. Also, repurposing of pipelines for CO₂ transport may be challenged by local rules and regulations, and interpretation of such. Thus, re-qualification of pipelines, especially for CO₂ transport in dense phase, is therefore not considered trivial and requires a careful evaluation but is nevertheless considered as fully possible in many cases.

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